AF2BIND

Predicting small-molecule binding sites using the pair representation of AlphaFold2

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Where could a small molecule bind?

Question
Can a protein structure predictor encode ligand-binding information even though it does not explicitly model small molecules?

- Homology transfer needs an already liganded relative.
- Pocket finders favor geometric cavities; shallow functional interfaces may be missed.
- Training a large binding-site network from scratch is difficult with sparse, redundant structural data.

Proposal: reuse a rich pretrained representation; train only a simple classifier.

Introduction; Results, Fig. 1.

The prediction target is a residue label

Inputs

Target amino-acid sequence and backbone coordinates; 20 disconnected amino-acid "baits."

No MSA and no identity or pose of the true small molecule are supplied.

Outputs

One score $P_i(\mathrm{bind})$ for each target residue i.

Spatial clustering subsequently turns high-scoring residues into sites.

Training definition

A residue is positive if a heavy atom lies within 5 Å of a ligand heavy atom; labels are augmented from closely related structures.

A binding score is neither an affinity estimate nor a predicted ligand pose.

Fig. 1; Methods: Dataset; Extracting AF2 features.

Why amino acids can act as chemical probes

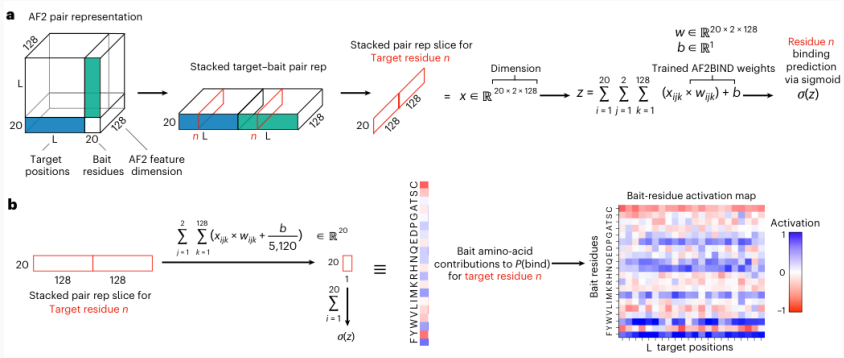
Many ligand functional groups resemble pieces of amino acids: aromatic rings, charged groups, amides and hydroxyls.

- Protein–protein contacts therefore provide a partial proxy for protein–ligand interactions.
- The authors give AF2 one of each canonical amino acid as an unplaced, disconnected probe.
- Their hypothesis: target–bait interactions reveal latent binding-site information in the pair representation.

Mechanism versus interpretation
"Relieving frustration" is an interpretation. A bait's final physical proximity to a residue need not track its binding score.

Introduction; Discussion; Extended Data Figs. 1–2. Amino-acid chemistry covers only part of ligand chemical space.

One AF2 pass, then a linear readout



Pair tensors in both directions connect each target residue to each bait; flatten them and apply logistic regression.

Reproduced from Fig. 2a–b, Gazizov et al., Nature Methods (2026).

What exactly becomes a feature?

For target residue i and bait type a, concatenate the two directed 128-dimensional pair embeddings:

$$x_{ia} = [z_{i,a}; z_{a,i}] \in \mathbb{R}^{256}, \qquad x_i = [x_{i1}; \ldots; x_{i20}] \in \mathbb{R}^{5120}.$$

After feature standardization using training-set statistics,

$$\ell_i = b + w^\top \tilde{x}_i, \qquad P_i(\mathrm{bind}) = \sigma(\ell_i) = \frac{1}{1 + e^{-\ell_i}}.$$

- Baits are separated with residue-index offsets of 50; no bait template.
- Target template masks side-chain dihedrals beyond Cβ.
- One forward pass, no recycles; typical 100–300-residue input: about 40 s on a T4 GPU.

Fig. 2; Methods: Extracting AF2 features; Runtime. Pair features are learned embeddings, not literal contact probabilities.

Running example: score one residue, recover a site

Illustrative numbers, not experimental data.

For residue 42, suppose the intercept is -1.0 and the 20 bait contributions sum to 2.4.

$$\ell_{42} = -1.0 + 2.4 = 1.4, \qquad P_{42}(\text{bind}) \approx 0.80.$$

- This residue passes the selected threshold of 0.28.
- Nearby passing residues can form a spatial cluster: a predicted site.
- To evaluate "recovery," suppose a reference ligand has 10 contact residues. If 7 appear among the top 10 predictions, recovery is 70%.

Recovery uses the known number of positives for evaluation; inference does not know that number.

Toy example derived from model equation; Methods: Assessment metric; site clustering.

Curation and splitting are central contributions

Dataset

PDB snapshot: March 2023. Single-chain complexes; 40–500 residues; curated crystallographic and ligand quality.

1,897 proteins; 2,708 ligands; 987 unique ligand residue names. Peptide/nucleic-acid complexes excluded.

Prevent easy transfer

Separate sets by sequence, global fold, evolutionary classification and pocket similarity.

Exclude test pockets with harmonic-mean TM-score > 0.6 to training pockets. Final test: 67 complexes.

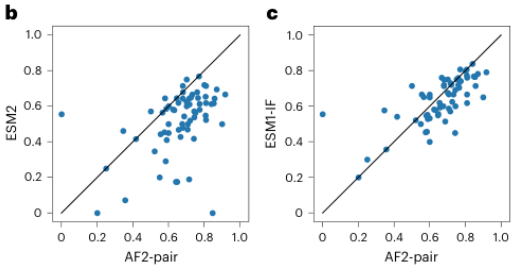
Ten cross-validation assignments; inverse structural-neighbor weighting; L2 regularization 0.03 limits overfitting.

Methods: Dataset; Robust splitting; Training. Splits concern AF2BIND supervision, not removal from AF2 pretraining.

Evidence: AF2 pair features lead single representations

a

Representations	Recovery	ROC AUC
AF2-single	0.454	0.840
ESM2 (sequence only)	0.523	0.872
ESM1-IF (structure only)	0.637	0.928
AF2-pair (AF2BIND)	0.662	0.936
AF2-pair + ESM2	0.675	0.940
AF2-pair + ESM1-IF	0.689	0.945
AF2-pair + ESM2 + ESM1-IF	0.690	0.945



- Mean recovery: AF2-pair 66%; ESM1-IF almost 64%; sequence-only ESM2 over 50%.
- Concatenating AF2-pair with ESM2 or ESM1-IF raises recovery to 68% or 69%.
- A common logistic readout and curated split isolate the usefulness of representations.

Fig. 3a–c reproduced above; Results and Discussion. Table: cross-validation; scatters: held-out test proteins.

Good discrimination does not settle calibration

Reported performance

ROC AUC: 0.936

Average precision: 0.728

Threshold $P(\mathrm{bind}) > 0.28$ selected using validation MCC/F1:
recall 67%, precision 63%, false-positive rate 4.3%.

How to read this

Recovery, AUC and threshold metrics answer different questions.

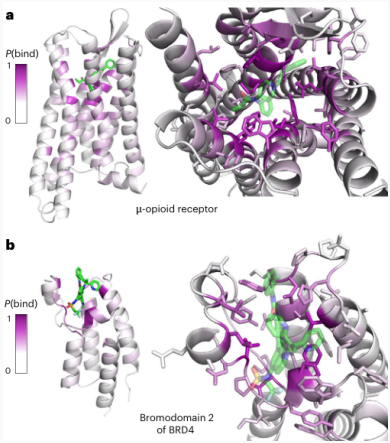
With only about 10% positive residues in training, precision and false positives matter for proteome screening.

Our statistical interpretation

The authors call threshold $\approx$ miss rate "well calibrated." This is not the usual test that predictions near p are positive with frequency p; reliability needs a separate assessment.

Results; Discussion; Extended Data Figs. 4–6. Values are reported summaries, not a new pooled evaluation.

Held-out families: an informative demonstration



Examples

Mu opioid receptor bound to fentanyl (8EF5); BRD4 second bromodomain (7RUH).

GPCRs and bromodomains were held out from classifier training and validation.

Scores localize known binding regions without the ligand as input.

These are examples of classifier generalization; they do not establish unseen AF2 pretraining structures.

Fig. 4a–b reproduced at left; Results: Exemplar performance. White to purple denotes score 0 to 1.

Bait attribution suggests compatible chemistry

Because the classifier is linear, the logit decomposes into bait-specific terms:

$$\ell_i = b + \sum_{a=1}^{20} c_{ia}, \qquad c_{ia} = w_a^\top \widetilde{x}_{ia}.$$

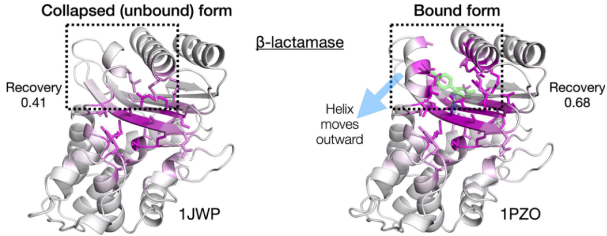
- For hydrophobic 2,3-oxidosqualene binding protein (4OMJ), aromatic baits W/F contribute strongly.
- For the polar substrate-binding kinase 2V2Z, polar Q/N contributions are more prominent.
- Across the dataset, bait combinations correlate with ligand hydrophobicity.

What interpretability buys
A chemically suggestive fingerprint of a site, rather than identification of a unique ligand or a causal binding mechanism.

Fig. 2b; Fig. 5a–d. Contributions add on the logit scale; they do not add as probabilities.

Apo structures: robustness has a boundary

Ten matched apo/holo pairs with Cα RMSD 1–3 Å show Spearman score correlations of 0.86–0.956; recovery is broadly similar but varies across pairs.



Cryptic β-lactamase site

A collapsed helix hides the apo pocket. Recovery: 0.41 apo versus 0.68 holo.

Global structural similarity cannot guarantee local pocket accessibility.

Supplementary Table 1; Extended Data Fig. 10 reproduced above. Larger-motion examples: Extended Data Fig. 9.

Scaling requires more than running on a long chain

1. Remove low-confidence segments ($\mathrm{pLDDT} < 50$); split structures with ≥ 300 residues into domain subgraphs and contacting domain pairs.
2. Run AF2BIND on the pieces; take each residue's maximum score across predictions. Suppress exposed-by-cutting interdomain contacts.
3. On the full structure, zero very buried residues; cluster scores > 0.28 using DBSCAN (6 Å heavy-atom distance; minimum samples 3).
4. Retain sites with at least 5 residues; split clusters larger than 50 residues.

Reason: long proteins dilute score magnitude. Domain handling changes the operational predictor and adds assumptions.

Methods: Analysis of large proteins; Binding-site identification; Supplementary Figs. 14–15.

The human proteome: many complementary hypotheses

AF2BIND catalogue

20,302 predicted sites in 13,686 proteins.

15,755 sites have no residue overlap with ligand annotations transferred by AlphaFill.

Comparison with P2Rank

9,732 AF2BIND sites distinct from P2Rank; 8,408 P2Rank sites distinct from AF2BIND.

Combined: nearly 29,000 sites in 15,026 proteins.

Site overlap means sharing at least one residue. Comparison uses site-level confidence filters; overlapping counts are not a simple partition because mappings can be many-to-one.

Interpretation

Novel predictions relative to two tools are hypotheses, not experimentally confirmed new binding sites.

Fig. 6a; Results: Prediction across the human proteome; Methods: comparison definitions.

Druggability and disease relevance: encouraging proxies

- Both AF2BIND and P2Rank sites have median SiteMap Dscore above 0.83, the paper's cited druggability threshold.
- AF2BIND sites are generally more exposed and less enclosed than P2Rank sites.
- In Morbid Map: sites predicted in 3,556 proteins; 411 have no shared site with either AlphaFill or P2Rank.

Ranking versus validation

AF2BIND's top-23 score has Spearman correlation 0.40 with Dscore (P2Rank probability: 0.31). Dscore is a computational geometry/chemistry proxy; it does not demonstrate affinity, selectivity or cellular activity.

Shallow peptide/RNA/DNA interfaces can score highly: useful opportunities, but also ambiguity about small-molecule ligandability.

Fig. 6b–c; Supplementary Figs. 3, 6, 8–10; Results: Quality; Morbid Map; Shallow sites.

Allosteric sites: small but relevant stress test

- Only 23 curated-dataset PDBs overlap an allosteric resource: about 1.2% of the dataset.
- In ligand-free AF2-predicted human structures, 27 of 39 allosteric sites are recovered (about 69%).
- A site counts as recovered if at least 3 true contact residues exceed 0.28; this is easier than recovering an entire site.

Limits on the claim

The curated test covers small, single-chain sites (Methods: 34 proteins, 39 sites). A separate nine-protein allosteric analysis includes seven training examples. Neither establishes broad allosteric prediction accuracy.

Supplementary Figs. 11–12; Methods: Prediction of allosteric sites. Fig. 12 describes 39 structures; use site denominator here.

Chemoproteomics provides an external check

212 liganded tryptic peptides, containing 271 cysteines, are cross-referenced against predicted sites in single-chain AF2 structures.

Prediction near experimentally liganded peptide	Matched peptides
AF2BIND	78 / 212
P2Rank	84 / 212
Either method	102 / 212

A match allows a predicted-site residue Cα within 12 Å of a liganded Cys side-chain heavy atom (Methods). Complementarity is supported; residue-level correctness and binding pose are not established by this generous proximity criterion.

Supplementary Fig. 13; Results: Chemoproteomics; Methods: proximal liganded Cys. Supplement caption instead says residue heavy atom.

What the evidence supports—and what remains open

Supported

Pretrained protein representations transfer to binding-residue prediction.

AF2 pair features plus a simple readout perform well on carefully separated families.

AF2BIND complements geometric pocket finding.

Open

How many unique proteome sites truly bind drug-like molecules?

Can a monomer model recover composite or collapsed cryptic sites?

How reliable are score probabilities under domain splitting and a different proteome?

Our next experiment: prospectively test matched unique/shared predictions, controlling for pocket size and exposure, with nonbinding controls and orthogonal binding assays.

Synthesis of Figs. 3–6, Extended Data Fig. 10, Supplementary Figs. 9–15; proposed experiment is our interpretation.

Discussion and practical use

1. Does bait attribution provide enough chemistry to prioritize fragments, or only a useful site descriptor?
2. Which benchmark would best separate representation quality from crystallographic-label and pretraining biases?
3. For a new target, would we combine AF2BIND and P2Rank before committing to experiments?

Take-home
A small supervised model can extract ligand-binding information from a protein structure model. Its strongest current use is generating complementary, experimentally testable site hypotheses.

Code and notebooks: `https://github.com/sokrypton/AF2BIND`

Human-proteome data: DOI 10.5281/zenodo.17683379.

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